Akademia Górniczo-Hutnicza
im. Stanisława Staszica w Krakowie

AGH University of Krakow

UAV Trajectory Planning

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Outline

1. What is 'Trajectory'?
2. Good UAV model property
3. Different UAV trajectories

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Part I

Motivation

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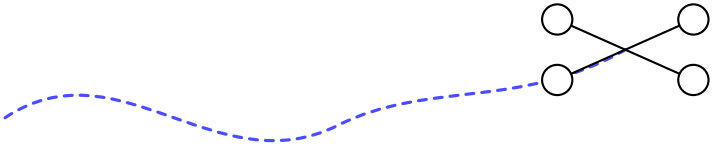
Trajectory – intuition



Intuitively – just a line that is drawn by a flying drone.

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Trajectory – formally

$$x = \begin{bmatrix} p(t) & v(t) & q(t) & \omega(t) \end{bmatrix}^T$$

Formally, the trajectory is the **state** of the UAV over time.

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Trajectory – theoretically

$$\begin{bmatrix} \dot{p} \\ \dot{v} \\ \dot{q} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \\ -g + \frac{1}{m} q \odot T \\ q \otimes \left(\frac{\omega_x}{2}\mathbf{i} + \frac{\omega_y}{2}\mathbf{j} + \frac{\omega_z}{2}\mathbf{k}\right) \\ J^{-1}(\tau - \omega \times J\omega) \end{bmatrix}$$

From theoretical point of view, trajectory is a **solution** of the state equation.

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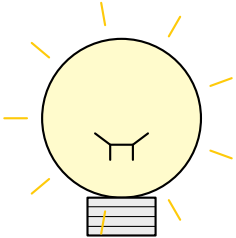
Clever solution?



If trajectory is a solution of the state equation, then we can just solve it for specific initial conditions and we will have a very good plan for the trajectory, right?

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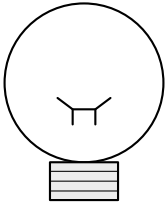
Clever solution? Well, not



- It's just a **model** – *sim2real* gap is a ~~bitch~~ big issue.

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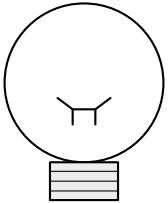

- It's just a **model** – *sim2real* gap is a ~~bitch~~ big issue.
- It is a **nonlinear** model – no analytical solution.

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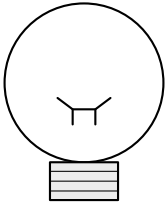

- It's just a **model** – *sim2real* gap is a ~~bitch~~ big issue.
- It is a **nonlinear** model – no analytical solution.
- This model has **inputs** – they need to be known somehow.

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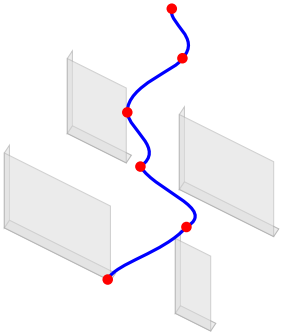
Typical solution

1. Determine waypoints & obstacles.
2. Define optimization criterion, like minimize time.
3. Solve the optimization problem.



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Part II

Differential flatness

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Formal definition

Flatness [M. Fliess et al. 1992]

A nonlinear system:

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}), \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{u} \in \mathbb{R}^m$$

is flat, if there exists an output:

$$\mathbf{y} = (y_1, ..., y_m)$$

that satisfies the following conditions:

1. $\mathbf{y} = \Phi(\mathbf{x}, \mathbf{u}, \dot{\mathbf{u}}, \cdots, \mathbf{u}^{(\alpha)})$
2. $\mathbf{x} = g(\mathbf{y}, \dot{\mathbf{y}}, \cdots, \mathbf{y}^{(\beta)}), \quad \mathbf{u} = h(\mathbf{y}, \dot{\mathbf{y}}, \cdots, \mathbf{y}^{(\beta)})$
3. $\psi(\mathbf{y}, \dot{\mathbf{y}}, \cdots, \mathbf{y}^{(\beta)}) = \mathbf{0}$ only if ψ is a null function (differential independence)

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Building intuition

Linear (LTI) systems
$\dot{x} = Ax + Bu$
Controllability (Kalman):
$\operatorname{rank}\left[B \;\; AB \;\; \cdots \;\; A^{n-1}B\right] = n$

generalization
(LTI: controllable $\Leftrightarrow$ flat)

Nonlinear systems
$\dot{x} = f(x, u)$
Differential flatness:
there exists an output y such that
$x = g(y, \dot{y}, \dots), \quad u = h(y, \dot{y}, \dots)$

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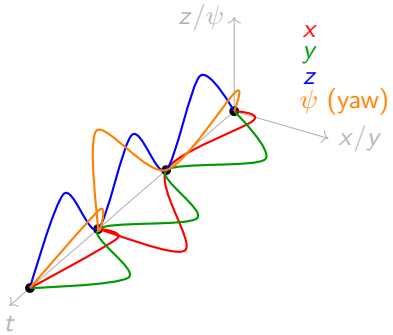
UAV flat outputs

UAV model has 4 inputs $(T, \tau_x, \tau_y, \tau_z)$, hence we are looking for 4 flat outputs. One of the possible solution is to use the **position** with **yaw** angle.



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UAV flat outputs – proof [Mellinger & Kumar 2011]

If

$$\sigma = \begin{bmatrix} x & y & z & \psi \end{bmatrix}$$

are *true* flat outputs, then we should be able to obtain from them (and their derivatives) the entire state vector:

$$x = \begin{bmatrix} p & v & q/R & \omega \end{bmatrix}^T$$

as well as control inputs:

$$u = \begin{bmatrix} T & \tau \end{bmatrix}^T$$

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UAV flat outputs – position & velocity

No problem with that since:

$$\mathbf{p} = \begin{bmatrix} x & y & z \end{bmatrix}$$

$$\mathbf{v} = \begin{bmatrix} \dot{x} & \dot{y} & \dot{z} \end{bmatrix}$$

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UAV flat outputs – total thrust

Based on the model second equation:

$$m\mathbf{a} = -mg\mathbf{z_W} + T\mathbf{z_B}$$

$$T\mathbf{z_B} = m\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} + g \end{bmatrix}$$

$$T\mathbf{z_B} = m\mathbf{a_g}$$

So the value of thrust is equal to

$$T = m\|\mathbf{a_g}\|$$

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UAV flat outputs – rotation

We can express the $\mathbf{z_B}$ in the world frame:

$$T\mathbf{z_B} = m\mathbf{a_g}$$
$$\mathbf{z_B} = \frac{\mathbf{a_g}}{\|\mathbf{a_g}\|}$$

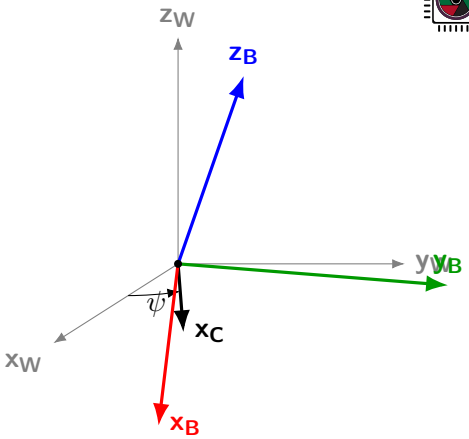
Then we determine the desired heading:

$$\mathbf{x_C} = \begin{bmatrix}\cos\psi & \sin\psi & 0\end{bmatrix}^T$$

and based on that we place the body frame:

$$\mathbf{y_B} = \frac{\mathbf{z_B} \times \mathbf{x_C}}{\|\mathbf{z_B} \times \mathbf{x_C}\|}$$
$$\mathbf{x_B} = \mathbf{y_B} \times \mathbf{z_B}$$

which gives us: $\mathbf{R} = \begin{bmatrix}\mathbf{x_B} & \mathbf{y_B} & \mathbf{z_B}\end{bmatrix}$



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UAV flat outputs – angular velocity (1)

Based – again – on the model second equation:

$$m\mathbf{a} = -mg\mathbf{z_W} + T\mathbf{z_B} \quad / \frac{d}{dt}$$

$$m\dot{\mathbf{a}} = \dot{T}\mathbf{z_B} + T\dot{\mathbf{z}}_\mathbf{B}$$

$$= \dot{T}\mathbf{z_B} + T(\omega \times \mathbf{z_B})$$

Now, by projecting it along $\mathbf{z_B}$:

$$m\dot{\mathbf{a}}\mathbf{z_B} = \dot{T}$$

By substitution to the previous equation:

$$m\dot{\mathbf{a}} = m(\dot{\mathbf{a}}\mathbf{z_B})\mathbf{z_B} + T(\omega \times \mathbf{z_B})$$

$$(\omega \times \mathbf{z_B}) = \frac{m}{T}(\dot{\mathbf{a}} - (\dot{\mathbf{a}}\mathbf{z_B})\mathbf{z_B}) = \mathbf{h}_\omega$$

Vector $\mathbf{h}_\omega$ can be viewed as the projection of the vector $\frac{m}{T}\dot{\mathbf{a}}$ onto the plane $\mathbf{x_B} - \mathbf{y_B}$.

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UAV flat outputs – angular velocity (2)

By cross product we have:

$$\mathbf{h}_\omega = \omega_y \mathbf{x_B} - \omega_x \mathbf{y_B}$$

Therefore:

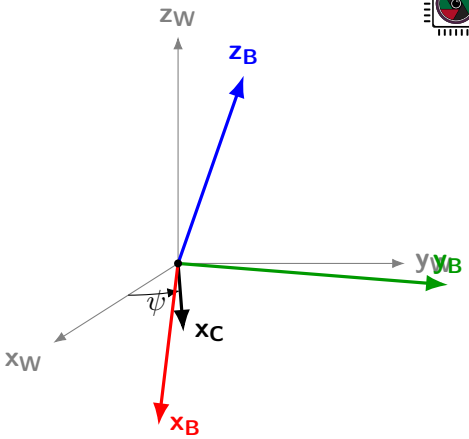
$$\omega_x = -\mathbf{h}_\omega \mathbf{y_B}$$
$$\omega_y = \mathbf{h}_\omega \mathbf{x_B}$$

To find ω_z we can simply note that:

$$\omega = \omega_{BC} + \omega_{CW}$$

and project it along $\mathbf{z_B}$:

$$\omega_\mathbf{z} = \omega_\mathbf{CW} \mathbf{z_B} = \dot{\psi} \mathbf{z_W} \mathbf{z_B}$$



UAV flat outputs – torque

To obtain torque we can use model last equation:

$$\dot{\omega} = J^{-1}(\tau - \omega \times J\omega)$$
$$\tau = J\dot{\omega} + \omega \times J\omega$$

So, we *only* need to find the angular acceleration $\dot{\omega}$. To do that we can use the same approach as with angular velocity, but take the second derivative of the force equation.

After that and *few more* calculation we will end up with the following formulas:

$$\dot{\omega}_x = \frac{m}{T}\ddot{a}_x - 2\frac{m}{T}\dot{a}_z\omega_y - \omega_x\omega_z$$
$$\dot{\omega}_y = \frac{m}{T}\ddot{a}_y + 2\frac{m}{T}\dot{a}_z\omega_x - \omega_y\omega_z$$
$$\dot{\omega}_z = \ddot{\psi}\mathbf{z_W}\mathbf{z_B}$$

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UAV flatness

Key takeaway

Thanks to the UAV model flatness, when planning trajectory it is enough to find only 4 differentiable functions: $x(t)$, $y(t)$, $z(t)$ and $\psi(t)$.

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Part III

Different ways of planning trajectory

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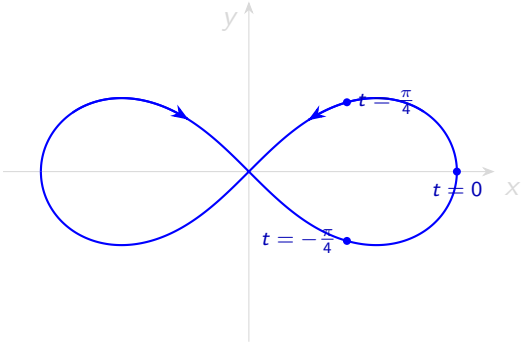
Predefined shapes

The simplest approach: just **parametrize** some curve and treat the parameter as a time.

Example: **lemniscate of Bernoulli**

$$x(t) = a\,\frac{\cos t}{1+\sin^2 t},$$
$$y(t) = a\,\frac{\sin t \cos t}{1+\sin^2 t}, \qquad t \in [-\pi, \pi].$$

- t is the curve parameter (often treated as time)
- arrows show the direction of increasing t



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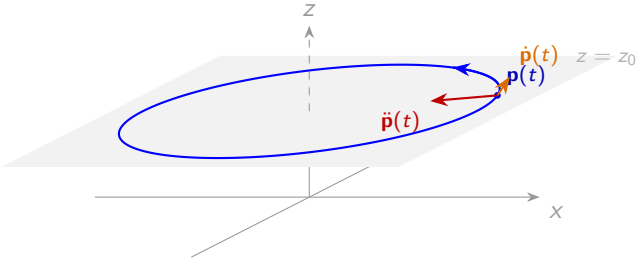
Case study: circular trajectory

A standard circle in the plane $z = z_0$ can be parametrized as

$$
\begin{aligned}
x(t) &= R\cos(\omega t), & \dot{x}(t) &= -R\omega\sin(\omega t), & \ddot{x}(t) &= -R\omega^2\cos(\omega t),\\
y(t) &= R\sin(\omega t), & \dot{y}(t) &= R\omega\cos(\omega t), & \ddot{y}(t) &= -R\omega^2\sin(\omega t),\\
z(t) &= z_0, & \dot{z}(t) &= 0, & \ddot{z}(t) &= 0.
\end{aligned}
$$

- R – radius, ω – angular speed, t – time
- speed magnitude is constant: $\|\dot{\mathbf{p}}(t)\| = R\omega$
- acceleration is centripetal: $\ddot{\mathbf{p}}(t) = -\omega^2\,\mathbf{p}_{xy}(t)$



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Circular trajectory – pros & cons

Pros:

- easy to find analytical derivatives up to any order,
- easy to limit maximum speed, acceleration, ...

Cons:

- not ideal as a base for spline,
- not optimal (even not sub-optimal) in any practical sense.

Good for **testing** and only for that.

Some possible modifications to make it more *interesting*:

- yaw angle control: $\psi(t) = \pi + \omega t$ (heading towards circle centre),
- vertical movement: $z(t) = kt$ (spiral trajectory).

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Polynomial trajectories

There are not many functions better than **polynomials**:

- they are continuos,

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Polynomial trajectories

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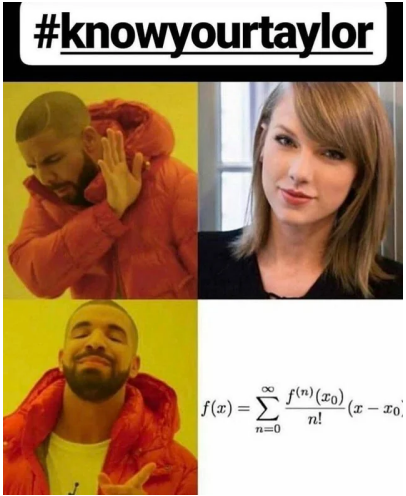
Polynomial trajectories

There are not many functions better than **polynomials**:

- they are continuos,
- they are easily differentiable,
- they can approximate other functions ☺.

So let's use polynomials for trajectory planning!



Case study: minimum snap trajectories [Mellinger & Kumar 2011]

Find trajectory:

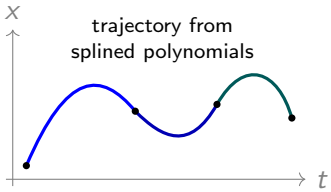
$$\sigma(t) = \begin{cases} \sum_{i=0}^{n} c_{0i}(t - T_0)^i & T_0 \leq t \leq T_1 \\ \sum_{i=0}^{n} c_{1i}(t - T_1)^i & T_1 \leq t \leq T_2 \\ \vdots & \\ \sum_{i=0}^{n} c_{(m-1)i}(t - T_{m-1})^i & T_{m-1} \leq t \leq T_m \end{cases}$$

that minimizes:

$$J = \int_0^{T_m} \left(\frac{d^4\sigma(t)}{dt^4}\right)^2 dt$$

with predefined time knots:

$$\sigma(t_j) = \sigma_j \quad \text{for} \quad j = 0, \cdots, m$$



Minimum snap solution

In order to make the problem easier to solve, we can use a polynomial algebra (ring) with time vectors as a base:

$$\sigma_j(t) = \begin{bmatrix} 1 & (t - T_j) & \cdots & (t - T_j)^7 \end{bmatrix} \begin{bmatrix} c_{j0} \\ c_{j1} \\ \vdots \\ c_{j7} \end{bmatrix}$$

With that we are able to find nice formula for k-th order derivative:

$${}^k\mathbf{t}[i] = \begin{cases} 0 & i < k \\ \frac{i!}{(i-k)!}(t - T_j)^{(i-k)} & i \geq k \end{cases}$$

Now our objective function transforms into:

$$\begin{aligned} J &= \sum_{j=0}^{m-1} \int_{T_j}^{T_{j+1}} ({}^4\mathbf{t}_j^T c_j)^T ({}^4\mathbf{t}_j^T c_j) dt \\ &= \sum_{j=0}^{m-1} \int_{T_j}^{T_{j+1}} c_j^T {}^4\mathbf{t}_j {}^4\mathbf{t}_j^T c_j dt \\ &= \sum_{j=0}^{m-1} c_j^T \left(\int_{T_j}^{T_{j+1}} {}^4\mathbf{t}_j {}^4\mathbf{t}_j^T dt \right) c_j \end{aligned}$$

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Minimum snap QP

Since

$$\int_{T_j}^{T_{j+1}} {}^4\mathbf{t}_j {}^4\mathbf{t}_j^T \, dt = M$$

is a constant matrix over time interval, we end my with a *well-known* Quadratic Programming (QP) formulation:

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2} c^T H c \\ \text{subject to} \quad & c_{lb} \leq c \leq c_{ub} \\ & a_{lb} \leq Ac \leq a_{ub} \end{aligned}$$

Further modifications:

- non-dimensional time,
- non-dimensional spatial dimension,
- time-knots optimization.

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